Section 4.2 — The Logarithm Function

Math 103: College Algebra

Translating Between Exponentials and Logs

Write each equation in logarithmic form.

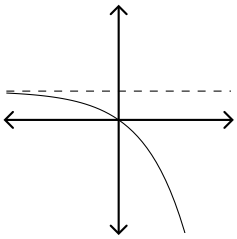
1. $5^4 = 625$
2. $7^{-4} = \frac{1}{2401}$
3. $\left(\frac{3}{4}\right)^3 = \frac{27}{64}$
4. $13^{1/2} = \sqrt{13}$

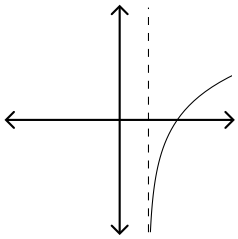
Write each equation in exponential form. Use your calculator to check that the new equation you have written is true.

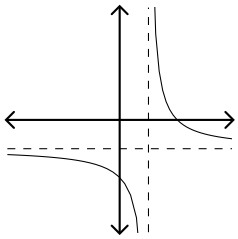
5. $\log_4 64 = 3$
6. $\log_3 \frac{1}{243} = -5$
7. $\log_{1/5} 25 = -2$
8. $\log_{25} 5 = \frac{1}{2}$

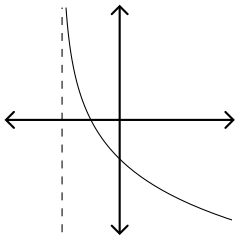
Graphs of Logarithms

Identify each graph below as that of an exponential function, a logarithmic function, or neither.

9.


10.


11.


12.


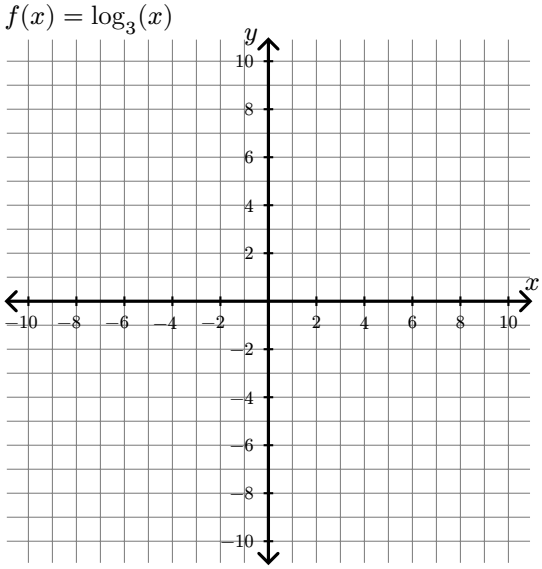
Of the two graphs above which are graphs of logarithm functions, choose the graph of each function below:

13. $f(x) = \log_2(x-1)$
14. $g(x) = \log_{3/5}(x+2)$

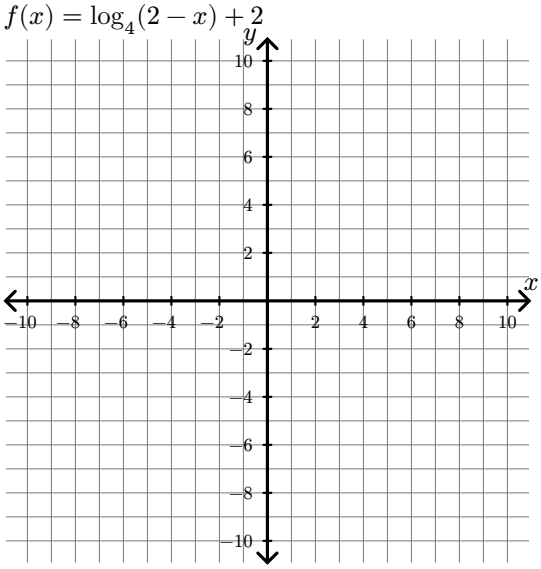
Graph a Logarithm

Sketch the graph of each logarithm function below in the space provided. Your graph should show the three points corresponding to outputs (not y values) -1, 0, and 1, as well as the asymptote. Do your best to estimate fractions.

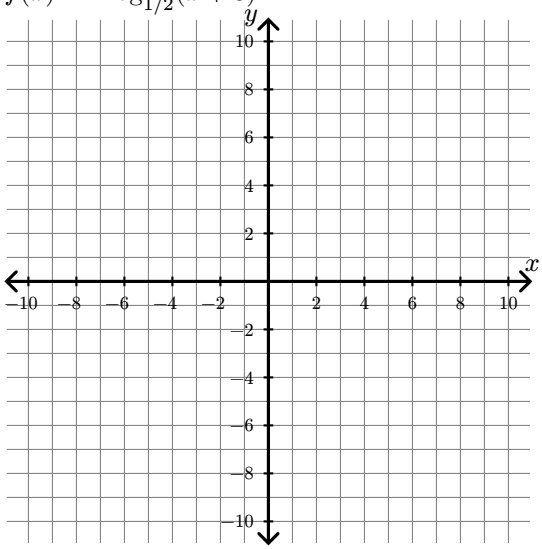
15. $f(x) = \log_3(x)$



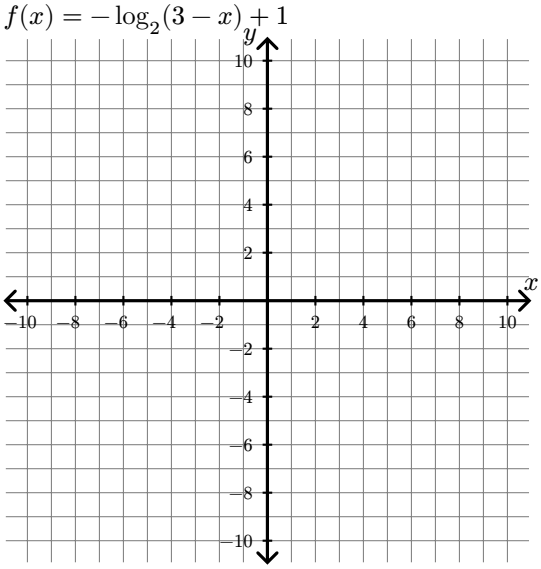
16. $f(x) = \log_4(2-x) + 2$



17. $f(x) = 2\log_{1/2}(x+3) - 1$



18. $f(x) = -\log_2(3-x) + 1$



Domain and Range of a Logarithm Function

For each function below, give the domain and range.

19. $h(x) = \log_{12}(x+3) + 5$

20. $k(x) = \log_{1/2}(5-x) - 1$

21. $u(x) = \frac{1}{3}\log_2(4x-12) + 7$

Common Logarithm

For each log below, rewrite in exponential form, then determine whether the sentence is true.

22. $\log 100 = 2$

23. $\log 0.001 = -2$

24. $\log 10000 = 6$

25. $\log 0.00001 = -5$

Calculate each log on your calculator. Round your answer to two decimal places if necessary. Check by rewriting in exponential form.

26. $\log 12$

27. $\log 13,225$

28. $\log 0.03$

29. $\log 1000$

Solving Equations Involving Logarithms — Basic

Solve each equation by rewriting in exponential notation. Round your answer to two decimal places, if necessary.

30. $\log_3 x = 2$

31. $\log_8 x = \frac{4}{3}$

32. $\log_x 25 = 2$

33. $\log_3(4x+1) = 4$

34. $\log_x \frac{1}{9} = -2$

35. $\log p = 4$

36. $\log_{x+2} 9 = 2$

37. $\log x = 1.1$